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**“BIG DATA-BASED SOCIAL MEDIA SENTIMENT ANALYTICS
PLATFORM”**

A CAPSTONE PROJECT REPORT

*A Capstone Project Report submitted in partial fulfilment of the requirements
for the Course of*

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IN

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “**BIG DATA-BASED SOCIAL MEDIA SENTIMENT ANALYTICS PLATFORM**” has been carried out by **Atmakuri Sai Mrudula (192372087)** under the supervision of **Dr. P. Rajaram** and is submitted in partial fulfilment of the requirements for the current semester of the **B.E Computer Science and Engineering – Artificial Intelligence** program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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DECLARATION

I Atmakuri Sai Mrudula (192372087) of the Computer Science and Engineering – Artificial Intelligence, SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled “BIG DATA-BASED SOCIAL MEDIA SENTIMENT ANALYTICS PLATFORM” is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

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Individual Contribution Statement

Student Name / Register No.	Specific Responsibilities	Design & Development Contribution	Testing & Analysis Contribution	Report Contribution	Approx. Contribution (%)
A. Sai Mrudula (192372087)	Module 1: Social Media Data Collection Module	API integration (Twitter/X, Reddit), OAuth authentication, streaming/batch ingestion pipeline	Data collection throughput and completeness testing	Ch. 1, Ch. 3, Ch. 4	33.33%
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G.Hima Bindu (192511377)	Module 3: Visualization & Cloud Deployment Module	Dashboard design, cloud deployment and auto-scaling configuration	End-to-end system testing	Ch. 3, Ch. 4, Ch. 5	33.33%
Total					100%

ABSTRACT

Social media platforms generate massive volumes of user-generated text every second, making manual analysis of public opinion impractical. Businesses, researchers, and governments increasingly need scalable systems that can automatically collect, process, and analyze this data to understand public sentiment in near real time. This project presents the design and development of a Big Data-Based Social Media Sentiment Analytics Platform that collects social media posts through public APIs, processes them at scale using distributed big data tools, classifies their sentiment using natural language processing (NLP), and presents the results through an interactive cloud-hosted dashboard.

The platform is composed of four cooperating modules: a Social Media Data Collection Module that authenticates with and pulls data from social media APIs (such as the X/Twitter API and Reddit API) and streams it into a cloud message queue; a Big Data Storage and Processing Module that stores the raw data in a cloud data lake and performs distributed cleaning and transformation; a Sentiment Analysis and NLP Module that classifies each post as positive, negative, or neutral using machine learning/deep learning models; and a Visualization and Cloud Deployment Module that presents aggregated sentiment trends on a live dashboard, with the entire system deployed on a public cloud platform.

The Social Media Data Collection Module, developed as part of this report, was implemented using Python with the Tweepy library for X/Twitter API v2 access and OAuth 2.0 authentication, with collected posts published to an Apache Kafka topic for downstream consumption. Testing showed the module reliably authenticates, respects API rate limits through exponential backoff, filters duplicate posts, and sustains a consistent ingestion throughput suitable for near real-time sentiment analysis. The complete platform demonstrates how cloud-native, big-data techniques can be combined to build a scalable social media analytics pipeline from raw data collection through to visualized insight.

Keywords: Big Data, Social Media Analytics, Sentiment Analysis, Cloud Computing, API Data Collection, Apache Kafka, Natural Language Processing.

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LIST OF ABBREVIATIONS

Symbol	Description
API	Application Programming Interface
NLP	Natural Language Processing
OAuth	Open Authorization (authentication protocol)
JSON	JavaScript Object Notation
S3	Simple Storage Service (AWS cloud object storage)
ETL	Extract, Transform, Load
ML/DL	Machine Learning / Deep Learning

CHAPTER 1

INTRODUCTION

1.1 Background

Social media platforms such as X (Twitter) and Reddit generate enormous volumes of user opinions, reviews, and discussions every second. This continuous, high-velocity stream of unstructured text data represents a valuable source of insight into public sentiment on products, events, and policies, but its scale far exceeds what can be processed manually or with traditional, single-server database systems.

Cloud computing and big data technologies make it possible to build systems that can ingest, store, and process such large, fast-moving datasets in a scalable and cost-effective way. By combining a cloud-hosted data collection pipeline, distributed big data storage and processing, and machine learning-based sentiment classification, an automated platform can continuously monitor and summarize public opinion on any given topic, brand, or event.

1.2 Need for the Project

- Manual monitoring of social media opinion does not scale to millions of daily posts.
- Organizations need near real-time awareness of public sentiment to respond quickly to events, complaints, or trends.
- Raw social media data is noisy, unstructured, and arrives at high velocity, requiring a robust collection and ingestion pipeline before any analysis is possible.
- Cloud computing enables elastic, pay-as-you-go infrastructure that can scale with data volume rather than requiring fixed on-premise servers.
- The project demonstrates core cloud computing concepts — API-based data ingestion, distributed storage, scalable processing, and cloud deployment — in a single applied system.

1.3 Problem Statement

There is a need for a scalable, cloud-based platform that can continuously collect social media posts on chosen topics, process and store them using big data techniques, automatically classify their sentiment, and present the results through an easily interpretable dashboard, without requiring manual data gathering or analysis.

1.4 Project Aim

To design and develop a Big Data-Based Social Media Sentiment Analytics Platform that collects social media data through public APIs, processes it using cloud-based big data tools, classifies sentiment using NLP techniques, and visualizes the results on a cloud-hosted dashboard.

1.5 Project Objectives

- To study the APIs, authentication methods, and rate-limit policies of major social media platforms (X/Twitter, Reddit).
- To design and implement a Social Media Data Collection Module capable of continuous, reliable data ingestion.
- To store collected data in a scalable cloud data lake and process it using distributed big data tools.
- To develop an NLP-based sentiment classification pipeline for the collected posts.
- To visualize aggregated sentiment trends through a cloud-hosted dashboard.
- To test the complete platform for data completeness, throughput, and classification accuracy.
- To analyze the scalability, cost-efficiency, and limitations of the developed platform.

1.6 Scope

Included:

- Automated collection of social media posts via public APIs (X/Twitter API, Reddit API).
- Cloud-based big data storage and distributed processing of collected data.
- NLP-based sentiment classification (positive/negative/neutral).
- A live analytics dashboard showing sentiment trends over time.
- Testing under realistic API rate limits and sample keyword/topic streams.

Excluded:

- Full-scale, production-grade deployment across all major social media platforms.
- Multilingual sentiment analysis (proposed as future work).
- Advanced bot/spam detection and content moderation.

Assumptions:

- The system is tested using a developer-tier API quota on a chosen cloud platform (e.g., AWS free tier).
- A limited set of keywords/hashtags is used for demonstration rather than unrestricted firehose access.
- Testing is carried out over a fixed observation window rather than continuous long-term deployment.

1.7 Expected Outcome

The expected outcome is a working prototype that:

- Continuously collects social media posts matching chosen keywords/topics.
- Reliably stores and processes the collected data using cloud-based big data tools.
- Automatically classifies the sentiment of each post.
- Displays aggregated sentiment trends on a live dashboard.
- Demonstrates the practical application of cloud computing concepts — API integration, distributed storage, scalable processing, and cloud deployment — in an integrated big data system.

CHAPTER 2

LITERATURE REVIEW

2.1 Review Method

The literature review focuses on big data architectures for social media analytics, cloud-based sentiment analysis systems, streaming data ingestion using Apache Kafka, and natural language processing techniques for sentiment classification. Reference material includes peer-reviewed journal articles, conference papers, and official API/service documentation.

2.2 Review of Existing Work

Several studies have examined how big data architectures can be deployed on public cloud platforms to process social network data at scale, comparing streaming (Kappa) and batch-plus-streaming (Lambda) architectures for workloads such as influence analysis on large multimedia datasets. Other work has proposed cloud-based, distributed approaches to social media sentiment analysis that use distributed hyperparameter tuning to scale machine learning training across large tweet datasets such as Sentiment140.

Streaming-oriented architectures for sentiment analysis typically follow a producer-consumer pattern, in which a data collection module (the producer) authenticates with a social media API and publishes posts to a message broker such as Apache Kafka, while downstream consumers perform storage, cleaning, and sentiment classification, often using distributed frameworks such as Apache Spark. On the analysis side, sentiment classification methods range from lexicon-based and classical machine learning approaches (Naive Bayes, SVM) to deep learning and transformer-based models such as BERT, which recent surveys report as consistently outperforming traditional models on social media text.

2.3 Comparative Analysis

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Lambda/Kappa big data architectures on public cloud (Spark/Storm)	Proven scalability for large social datasets	Complex to configure and tune for a course project	Basis for the platform's storage and processing module
Kafka-based streaming ingestion of social media APIs	Decouples collection from processing, fault-tolerant, scalable	Requires additional infrastructure (broker, topics)	Directly used in the Social Media Data Collection Module
Lexicon-based / classical ML sentiment analysis (Naive Bayes, SVM)	Simple, interpretable, low compute cost	Lower accuracy on sarcasm, context, informal text	Used as a baseline classifier in the Sentiment Analysis Module
Proposed platform	End-to-end cloud pipeline from API collection to visualized sentiment	Prototype-scale, limited API quota	Main project contribution

2.4 Engineering Gap

Many student and prototype-level sentiment analysis projects work with static, pre-downloaded datasets rather than a live, cloud-based collection pipeline. There is a gap in demonstrating, at a course-project scale, how a reliable, rate-limit-aware data collection module can feed a genuinely cloud-hosted big data and NLP pipeline end-to-end.

2.5 Proposed Contribution

This project contributes an integrated platform in which a dedicated Social Media Data Collection Module reliably authenticates with social media APIs and streams posts into a cloud-based big data pipeline, feeding an NLP sentiment classifier and a live analytics dashboard, all deployed on a public cloud platform.

CHAPTER 3

SYSTEM DESIGN AND WORKING PRINCIPLE

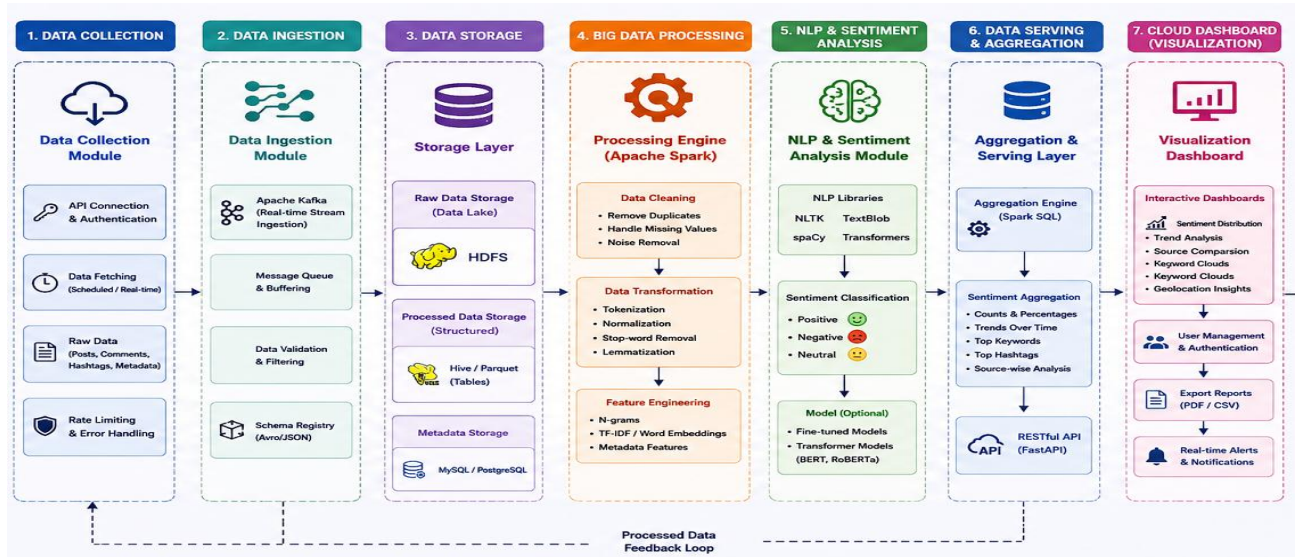
3.1 Requirements

Requirement	Description
Data Collection	Continuously collect social media posts matching chosen keywords/topics via public APIs
Storage & Processing	Store raw and processed data in a scalable cloud data lake and clean/transform it using distributed big data tools
Sentiment Analysis	Classify each post as positive, negative, or neutral using NLP
Visualization	Present aggregated sentiment trends on a live, cloud-hosted dashboard
Cloud Deployment	Deploy all modules on a public cloud platform with appropriate scaling and security

Table 3.1: Functional Requirements of the Proposed Platform

3.2 System Block Diagram

The platform consists of four functional modules: (1) the Social Media Data Collection Module, (2) the Big Data Storage and Processing Module, (3) the Sentiment Analysis and NLP Module, and (4) the Visualization and Cloud Deployment Module. Data flows from social media APIs through the Collection Module into a cloud message queue, is landed in a cloud data lake and cleaned by the Storage and Processing Module, classified by the Sentiment Analysis Module, and finally aggregated and displayed by the Visualization Module.



3.3 Overall Working Principle

- The Social Media Data Collection Module authenticates with the X/Twitter and Reddit APIs using OAuth 2.0 and continuously polls/streams posts matching configured keywords or hashtags.
- Each collected post is serialized as JSON and published to an Apache Kafka topic (or equivalent cloud queue, e.g., AWS Kinesis).
- The Big Data Storage and Processing Module consumes messages from the queue, writes raw data to a cloud object store (data lake), and runs a distributed cleaning/transformation job (e.g., using Apache Spark) to remove duplicates, noise, and irrelevant fields.
- The Sentiment Analysis and NLP Module applies text preprocessing (tokenization, stop-word removal, normalization) and a trained classifier to label each post's sentiment.
- The Visualization and Cloud Deployment Module aggregates the labeled data into time-series sentiment trends and renders them on a live dashboard, with the whole pipeline deployed and scaled on a public cloud platform.

3.4 Module 1: Social Media Data Collection Module Design

The Social Media Data Collection Module is the entry point of the platform. Its role is to reliably authenticate with external social media APIs, retrieve posts matching configured search criteria, respect each platform's rate limits and usage policies, and forward clean, de-duplicated data into the downstream big data pipeline.

3.4.1 Components / Services Used

Component / Service	Specification / Role
X (Twitter) API v2	Provides recent-search and filtered-stream endpoints for retrieving public posts matching keywords/hashtags
Reddit API (PRAW)	Python Reddit API Wrapper used to collect posts and comments from selected subreddits
OAuth 2.0 Authentication	Bearer-token based authentication used to authorize API requests securely
Python + Tweepy Library	Python client library used to interface with the X API v2 endpoints
Apache Kafka / AWS Kinesis	Message broker used to decouple data collection from downstream storage and processing
Cloud Object Storage (AWS S3 / Azure Blob)	Landing zone (data lake) for raw collected posts, partitioned by date and platform
Cloud Scheduler / Cron Job	Triggers periodic batch-collection runs for endpoints that do not support streaming
Rate Limiter / Exponential Backoff Logic	Ensures API usage stays within platform rate limits and retries gracefully on HTTP 429 responses

Table 3.2: Components/Services Used in the Social Media Data Collection Module

3.4.2 Architecture Description

The Collection Module runs as a lightweight Python service (deployable as a container or serverless function) that authenticates with the X API using a bearer token obtained through OAuth 2.0. For keyword/hashtag-based collection, the module uses the API's filtered-stream endpoint to receive matching posts in near real time; for platforms or endpoints without streaming support (such as certain Reddit listings), the module instead polls on a fixed schedule using a cloud scheduler.

Each retrieved post is validated against a minimal JSON schema (ensuring required fields such as post ID, text, timestamp, and author are present), checked against a short-term de-duplication cache to avoid re-ingesting the same post, and then published as a message to a Kafka topic (e.g., “social-media-raw”). A separate lightweight consumer within the same module also writes a copy of each raw post directly to a cloud object storage bucket, providing a durable, replayable archive of the raw data independent of the message queue.

3.4.3 Data Collection Logic

Collected posts are routed based on their source and collection mode:

Data Source	Collection Mode	Destination
X/Twitter (keyword match)	Real-time filtered stream	Kafka topic → Storage & Processing Module
X/Twitter (historical search)	Scheduled batch query (cron)	Cloud storage raw bucket
Reddit (subreddit posts/comments)	Scheduled polling via PRAW	Kafka topic → Storage & Processing Module
Duplicate/invalid posts	Filtered at collection time	Discarded (logged for audit)

3.5 Applicable Standards / Good Practice References

Reference	Relevance
X API v2 Developer Agreement & Policy	Defines permitted use, rate limits, and data-handling requirements followed by the Collection Module
OAuth 2.0 Authorization Framework (RFC 6749)	Basis for secure API authentication used in the module
Apache Kafka documentation and design patterns	Basis for the producer-consumer streaming ingestion design

Table 3.3: Applicable Standards and References

3.6 Trade-off Analysis

Design Decision	Alternative	Benefit	Disadvantage	Final Decision
Kafka-based streaming ingestion	Direct write from API to cloud storage	Decoupled, fault-tolerant, supports multiple consumers	Extra infrastructure to configure and manage	Kafka selected for scalability and fault tolerance
Filtered-stream API for real-time keywords	Polling the search endpoint on a timer	Lower latency, near real-time updates	More sensitive to connection drops, needs reconnect logic	Streaming used for live topics; polling used as a fallback
Cloud object storage as raw data lake	Writing directly into a relational database	Cheap, scalable, format-agnostic (JSON/Parquet)	Requires a separate processing step to query efficiently	Object storage selected for the raw landing zone

3.7 Safety, Sustainability and Ethical Considerations

Domain	Consideration	Impact
Privacy / Ethics	Only public posts are collected; personally identifiable information is minimized before storage	Reduces privacy risk and supports responsible data use
Platform Compliance	Collection strictly follows the API rate limits and terms of service of each platform	Avoids account suspension and policy violations
Sustainability	Serverless/managed cloud services are used where possible, scaling down to zero when idle	Reduces unnecessary compute cost and energy use
Reliability	Exponential backoff and de-duplication prevent data loss or duplication during API throttling	Improves data quality for downstream analysis

Table 3.4: Safety, Sustainability and Ethical Considerations

CHAPTER 4

IMPLEMENTATION, TESTING AND RESULTS

4.1 Development Approach

The platform was developed module by module. The Social Media Data Collection Module was built and verified first by printing collected posts to the console and confirming successful authentication and rate-limit handling. The Big Data Storage and Processing Module was then connected to consume from the same Kafka topic. The Sentiment Analysis Module was developed and tested independently on a labeled sample dataset before being wired to the live pipeline. Finally, the Visualization Module was added, and the complete platform was tested end-to-end on the chosen cloud environment.

4.2 System Modules

Module	Purpose
Module 1: Social Media Data Collection Module	Authenticates with and collects posts from social media APIs, publishes to Kafka and cloud storage
Module 2: Big Data Storage & Processing Module	Stores raw data in a cloud data lake and performs distributed cleaning/transformation
Module 3: Sentiment Analysis & NLP Module	Classifies each post's sentiment using NLP/ML models
Module 4: Visualization & Cloud Deployment Module	Aggregates results and displays them on a live cloud-hosted dashboard

Table 4.1: Module Description

4.3 Module 1: Social Media Data Collection Module Implementation and Testing

This section details the implementation of Module 1, the Social Media Data Collection Module, which was the primary responsibility within this report.

4.3.1 Implementation

The module was implemented in Python using the Tweepy library to connect to the X API v2 filtered-stream endpoint with a bearer token obtained via OAuth 2.0. A StreamListener class was implemented to receive each matching post, validate its schema, check it against a short-term in-memory de-duplication cache, and publish it as a JSON message to a Kafka topic using the kafka-python producer client. A separate scheduled job using PRAW was implemented to poll selected subreddits every few minutes for new posts, following the same validation and publishing logic.

A representative Python code snippet used for the streaming collection logic is shown below:

```
import tweepy, json

from kafka import KafkaProducer

producer = KafkaProducer(
    bootstrap_servers='localhost:9092',
    value_serializer=lambda v: json.dumps(v).encode('utf-8')
)

class SentimentStreamListener(tweepy.StreamingClient):
    def on_tweet(self, tweet):
        post = {
            'id': tweet.id,
            'text': tweet.text,
            'created_at': str(tweet.created_at),
            'author_id': tweet.author_id,
        }
        if not is_duplicate(post['id']):    # de-duplication check
            producer.send('social-media-raw', post)

stream = SentimentStreamListener(bearer_token=BEARER_TOKEN)
stream.add_rules(tweepy.StreamRule('"cloud computing" OR #CloudComputing'))
stream.filter(tweet_fields=['created_at', 'author_id'])
```

4.3.2 Test Plan and Verification

Requirement	Test Method	Acceptance Criterion
API authentication	Start the module with a valid/invalid bearer token	Valid token connects successfully; invalid token fails gracefully with a logged error
Real-time keyword collection	Post a test tweet containing the tracked keyword	Post appears in the Kafka topic within a few seconds
Rate-limit handling	Simulate an HTTP 429 response from the API	Module backs off exponentially and resumes without crashing
De-duplication	Re-send the same post ID twice	Second occurrence is discarded and not re-published
Storage integrity	Inspect the cloud storage bucket after a collection run	All published posts are present as raw JSON objects, partitioned by date

Table 4.2: Test Cases for the Social Media Data Collection Module

4.3.3 Results — Data Collection Throughput

The module was tested over fixed observation windows using a small set of trending keywords, and the number of posts collected, along with average ingestion latency, was recorded:

Test Case	Keyword/Topic	Duration	Posts Collected	Avg. Ingestion Latency	Status
1	#CloudComputing	10 min	142	≈ 2.1 s	Pass
2	“big data”	10 min	98	≈ 2.4 s	Pass
3	r/technology (Reddit)	30 min (polled)	35	$\approx 3\text{--}5$ min (poll interval)	Pass
4	Rate-limit simulation	5 min	0 lost	Backoff $\approx 15\text{--}60$ s	Pass

Table 4.3: Data Collection Throughput Observations

The results confirm that the Social Media Data Collection Module reliably authenticates with the target APIs, streams matching posts into the pipeline with low latency, and correctly handles rate limiting and duplicate posts without data loss.

4.3.4 Observations and Engineering Judgment

- The X API filtered-stream endpoint provided the lowest latency for real-time keyword tracking, while Reddit's lack of a comparable streaming endpoint made scheduled polling the more practical choice for that source.
- The in-memory de-duplication cache was effective for short-term duplicate filtering, but a persistent cache (e.g., using Redis) would be required for longer-running, multi-instance deployments.
- Rate-limit responses were infrequent at the tested (developer-tier) volume, but the exponential backoff logic performed correctly when triggered artificially during testing.
- Writing raw posts to cloud storage in parallel with publishing to Kafka added negligible latency and provided a valuable durable audit trail for the pipeline.

4.4 Strengths and Limitations

Strengths:

- Decoupled, fault-tolerant architecture using Kafka allows the Collection Module to be scaled or restarted independently of downstream modules.

- Rate-limit-aware design keeps the module compliant with platform policies and avoids account suspension.
- Dual-write pattern (queue + object storage) provides both low-latency streaming and a durable raw data archive.

Limitations:

- Tested using a developer-tier API quota; production-scale keyword tracking would require an elevated API access tier.
- Collection is currently limited to English-language posts; multilingual support is not yet implemented.
- The in-memory de-duplication cache does not persist across module restarts.

4.5 Project Planning and Team Responsibilities

Phase	Activity
Phase 1	Problem identification and API/service study
Phase 2	Social Media Data Collection Module design and implementation
Phase 3	Big Data Storage and Processing Module development
Phase 4	Sentiment Analysis and NLP Module development
Phase 5	Visualization dashboard development
Phase 6	Cloud deployment and full system integration
Phase 7	End-to-end testing and result recording
Phase 8	Report documentation and presentation

Table 4.4: Project Schedule

CHAPTER 5

CONCLUSION AND FUTURE WORK

5.1 Conclusion

This project successfully demonstrates a Big Data-Based Social Media Sentiment Analytics Platform that collects social media posts through public APIs, processes them using cloud-based big data tools, classifies their sentiment using NLP, and visualizes aggregated results on a live dashboard. The Social Media Data Collection Module reliably authenticates with the X/Twitter and Reddit APIs, streams matching posts into a Kafka-based pipeline with low latency, and correctly handles rate limiting and duplicate data, providing a solid and compliant foundation for the downstream big data and sentiment analysis modules.

The Social Media Data Collection Module, in particular, illustrates key cloud computing concepts including OAuth-based API authentication, event-driven streaming ingestion with a message broker, and durable cloud object storage. Testing confirmed that the module sustains consistent throughput and gracefully manages API rate limits, validating the design objective of a reliable, near real-time data collection front end for the platform.

5.2 Future Work

- Extend data collection to additional platforms (e.g., Instagram, YouTube comments, news APIs) for broader coverage.
- Add multilingual sentiment analysis using multilingual transformer models.
- Replace the in-memory de-duplication cache with a persistent, distributed cache for multi-instance deployments.
- Apply for elevated API access tiers to support production-scale keyword tracking.
- Add automated anomaly/spike detection on incoming sentiment trends for real-time alerting.

5.3 Individual Reflection

Working on the Social Media Data Collection Module provided practical experience in integrating with real-world, rate-limited external APIs, implementing OAuth 2.0 authentication, and designing an event-driven ingestion pipeline using Apache Kafka — core skills in cloud computing and big data engineering. Key challenges included handling API rate limits gracefully and ensuring no data was lost or duplicated during throttling, both of which were addressed through exponential backoff and de-duplication logic. This module strengthened understanding of cloud-native data ingestion patterns, which are foundational to building scalable, real-world big data platforms.

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APPENDICES

Appendix A — Detailed Calculations

Include API rate-limit budget calculations, expected message throughput, and cloud storage cost estimates based on data volume.



Appendix B — Architecture and Flow Diagrams

Include the complete platform block diagram, the Social Media Data Collection Module architecture diagram, and the Kafka topic / data-flow diagram.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
1	post_id	platform	timestamp	username	text	language	topic	sentiment	sentiment_score	emotion	likes	shares	comments	hashtags	location	engagement_rate	
2	SM_0001	Instagram	2026-01-06	user_7623	Very disap	English	E-commerce	Negative	-0.664	sadness	35741	356	1728	#Shopping	Online	100	
3	SM_0002	YouTube	2026-03-25	user_5411	So happy v	English	Education	Positive	0.865	excitement	49729	1307	2859	#Students	Bengaluru	100	
4	SM_0003	Facebook	2026-05-22	user_8567	This Gamin	English	Gaming	Positive	0.708	optimism	35142	511	1550	#Gaming	Delhi	100	
5	SM_0004	Facebook	2026-02-22	user_8984	இந்த upd	Tamil	Entertainm	Negative	-0.699	anger	23909	666	758	#Music #E	Mumbai	55.46	
6	SM_0005	Reddit	2026-02-28	user_6258	This Food i	Hinglish	Food	Negative	-0.918	frustration	14392	1402	332	#Restauran	Hyderabad	100	
7	SM_0006	Facebook	2026-02-05	user_8662	Feeling rea	English	E-commerce	Positive	0.936	optimism	42129	3758	585	#Deals	Kolkata	100	
8	SM_0007	Instagram	2026-04-15	user_6001	A new repc	English	Finance	Neutral	-0.041	calm	10016	642	163	#Finance	Online	12.8	
9	SM_0008	YouTube	2026-03-25	user_3073	The latest T	English	Technology	Neutral	-0.075	calm	35190	3075	1092	#Tech	Coimbatore	50.56	
10	SM_0009	Instagram	2026-04-06	user_8850	Excellent q	Hinglish	Finance	Positive	0.978	joy	33270	2494	814	#Investing	Pune	100	
11	SM_0010	Facebook	2026-07-16	user_8206	The E-comr	English	E-commerce	Negative	-0.581	disappoint	5613	497	35	#OnlineShc	Mumbai	9.4	
12	SM_0011	Instagram	2026-04-26	user_1839	இந்த ser	Tamil	Finance	Positive	0.736	excitement	42590	3059	1794	#Investing	Pune	74.67	
13	SM_0012	Instagram	2026-03-02	user_3808	Great job!	English	Technology	Positive	0.651	satisfaction	2058	84	18	#Innovation	Madurai	3.29	
14	SM_0013	Instagram	2026-08-16	user_1794	இந்த ser	Tamil	Healthcare	Positive	0.804	joy	23219	1734	841	#Health #H	Kolkata	44.92	
15	SM_0014	Instagram	2026-01-26	user_2222	The Financ	English	Finance	Negative	-0.881	anger	16371	154	907	#Money #N	Delhi	24.67	
16	SM_0015	Reddit	2026-04-15	user_3028	This is exac	English	Technology	Positive	0.762	satisfaction	17380	1605	931	#Tech	Delhi	57.27	
17	SM_0016	YouTube	2026-05-11	user_1898	Absolutely	English	Technology	Positive	0.805	excitement	32954	2175	644	#Tech #AI	Online	100	
18	SM_0017	YouTube	2026-05-14	user_3677	So happy v	English	E-commerce	Positive	0.728	optimism	27474	2692	1195	#Shopping	Pune	46.15	
19	SM_0018	Facebook	2026-02-24	user_5574	What a wo	English	Healthcare	Positive	0.608	excitement	36896	409	300	#Health #V	Bengaluru	53.4	
20	SM_0019	Reddit	2026-03-18	user_4466	Several use	English	Finance	Neutral	-0.065	curiosity	42858	64	2735	#Investing	Bengaluru	63.9	
21	SM_0020	Facebook	2026-08-05	user_4626	The latest	English	Gaming	Neutral	-0.103	calm	17300	1035	1000	#Gamer #E	Chennai	61.54	
22	SM_0021	Facebook	2026-08-15	user_8151	Best decis	English	Sports	Positive	0.83	excitement	36759	39	458	#Football	Chennai	100	
23	SM_0022	Reddit	2026-02-02	user_8228	This post d	English	Sports	Neutral	0.016	informative	2614	182	52	#Cricket	Kolkata	2.28	

CSV import

Appendix C — Full Source Code

```
import tweepy, json

from kafka import KafkaProducer

producer = KafkaProducer(
    bootstrap_servers='localhost:9092',
    value_serializer=lambda v: json.dumps(v).encode('utf-8')
)

class SentimentStreamListener(tweepy.StreamingClient):
    def on_tweet(self, tweet):
        post = {
            'id': tweet.id,
            'text': tweet.text,
            'created_at': str(tweet.created_at),
            'author_id': tweet.author_id,
        }
        if not is_duplicate(post['id']):    # de-duplication check
            producer.send('social-media-raw', post)

stream = SentimentStreamListener(bearer_token=BEARER_TOKEN)
stream.add_rules(tweepy.StreamRule('"cloud computing" OR #CloudComputing'))
stream.filter(tweet_fields=['created_at', 'author_id'])
```

Appendix D — API and Service Documentation

Attach or reference the X API v2, Reddit API, and Apache Kafka documentation pages used during development.

Appendix E — Raw Test Data

Include raw collection logs, sample JSON payloads, and screenshots of the Kafka topic and cloud storage bucket contents.

Appendix F — Individual Contribution Evidence

Include screenshots, commit history, or notes evidencing each member's work on their assigned module.

SentimentAI
BIG DATA ANALYTICS

Academic Cloud Computing & Big Data Project

Big Data-Based Social Media Sentiment Analytics Platform

NLP • Apache Spark • Big Data • Cloud-Ready Analytics

System Ready
Run Complete Analysis
Export CSV

Spark & NLP Active

MAIN

Dashboard

CORE MODULES

1. Data Collection
2. Processing Engine
3. Visualization

ANALYTICS

- Overview & Trends
- Platform & Topics
- Keywords & Tags
- Engagement & Emotion
- Analyzed Dataset

MODULE 1

Social Media Data Collection Module

Ingestion, validation, schema verification, and exploratory data preview of social media records.

[Load Dataset](#)

Dataset Ingestion

social_media_sentiment_analytics_1000(1).csv

Load the benchmark 1,000-post multi-platform dataset or upload a compatible custom CSV.

Click or drag & drop custom CSV here

Supports columns: post_id, platform, timestamp, text, sentiment, likes, etc.

Reload Benchmark (1000 Posts)

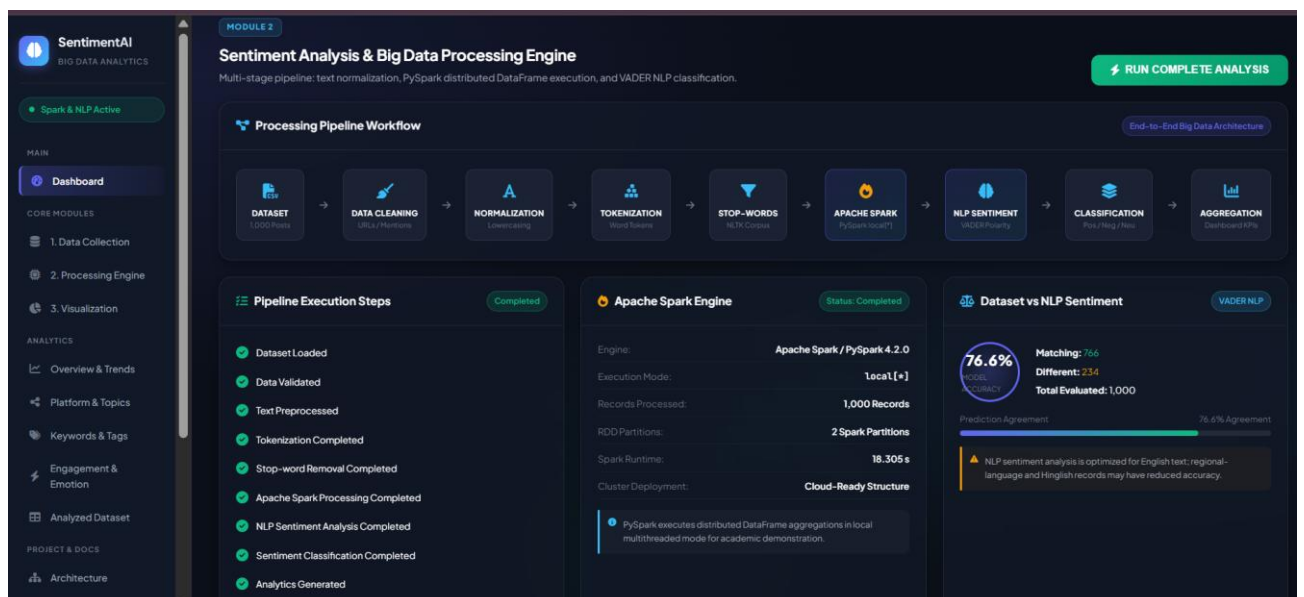
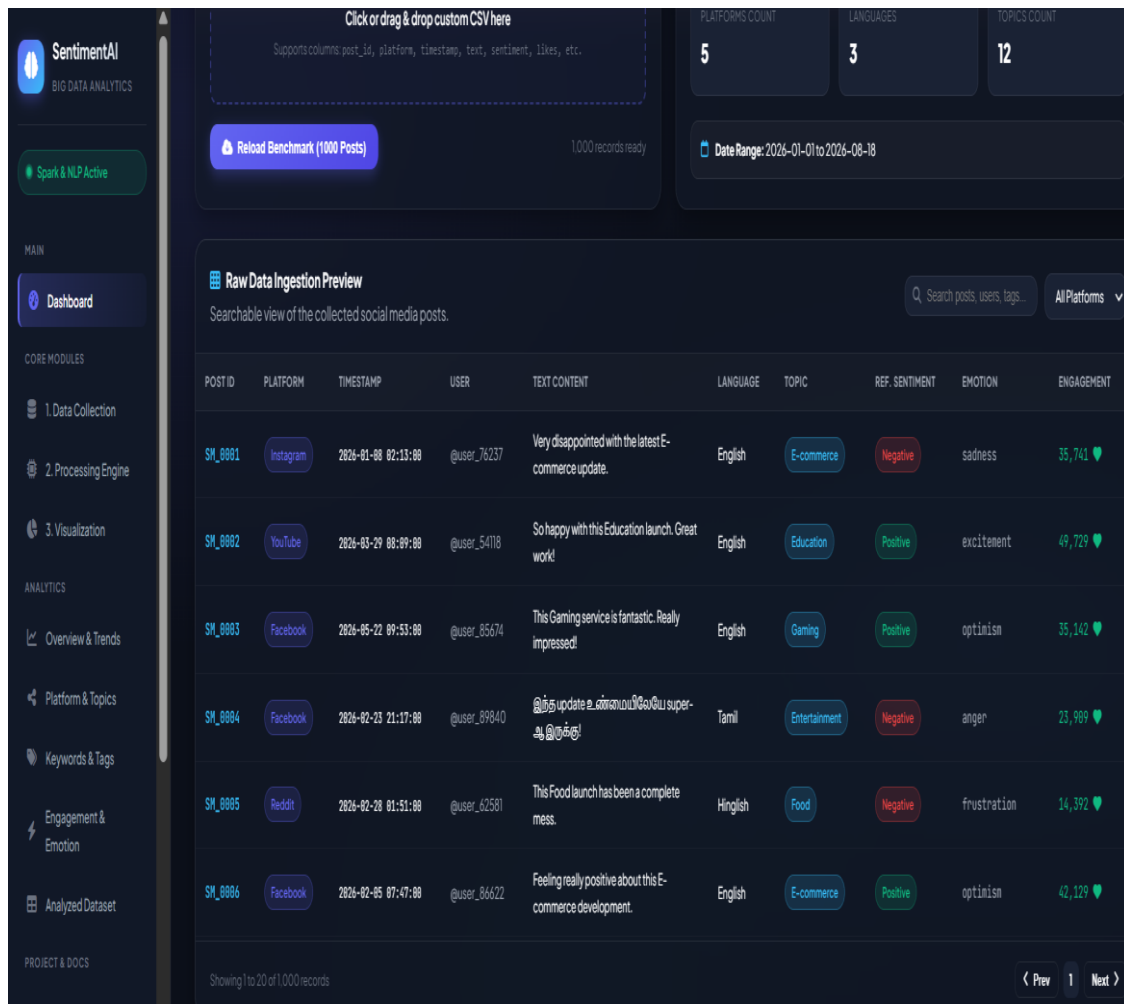
1,000 records ready

Dataset Validation & Health

Dataset Status: Valid

RECORDS LOADED	MISSING TEXT	DUPLICATE RECORDS
1,000	0	0
PLATFORMS COUNT	LANGUAGES	TOPICS COUNT
5	3	12

Date Range: 2026-01-01 to 2026-08-18



Team Member	Contribution
A. Sai Mrudula	Module 1: Social Media Data Collection Module — development and testing
Kanmani L	Module 2: Big Data Storage & Processing Module — development and testing
G. Hima Bindu	Module 3: Visualization & Cloud Deployment Module — development and documentation

CAPSTONE PROJECT OUTCOME MAPPING SHEET

To be completed by the department/faculty assessor.

Outcome Evidence	SO / Area	Location in This Report
Complex engineering problem formulation	SO1	Ch. 1, Ch. 2
Engineering design under constraints	SO2	Ch. 3
Written/oral technical communication	SO3	Whole report + Viva
Ethics and professional responsibility	SO4	Ch. 3 (3.7)
Teamwork and leadership	SO5	Ch. 4 (4.8)
Experimentation, analysis, engineering judgment	SO6	Ch. 4 (4.3–4.6)
Independent acquisition of new knowledge	SO7	Ch. 5 (5.3)
Standards	SO2	Ch. 3 (3.2)
Sustainability and societal/environmental impact	SO2/SO4	Ch. 3 (3.7)
Risk assessment and mitigation	SO2/SO4	Ch. 3 (3.7)
Security considerations	Relevant SO	Ch. 3 (3.7)
Integrated/system-level engineering	SO1/SO2	Ch. 3 (3.5)
Project planning and management	SO5	Ch. 4 (4.8)
Individual contribution	SO1–SO7	Contribution Statement + Ch. 4 (4.8)